Retest · Algebra I Module 2 Topic 1

Answer Key · Forms A, B, C (teacher copy)

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1 point each. Item 1 matches the original retest item 1 on every form, item 2 matches item 2, and so on.

Item	Question	Answer	Work
A1	What is the point of the line given by the point-slope equation $y - 5 = \frac{2}{3}(x + 1)$? Enter your answer as an ordered pair.	**$(-1, 5)$**	$y - y_1 = m(x - x_1)$: $x_1 = -1$, $y_1 = 5$.
A2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y - 2 = 3(x - 4)$**	**$y = 3x - 10$**	$y - 2 = 3x - 12 \rightarrow y = 3x - 10$
A3	What is the equation in slope-intercept form of a line with a slope of $\frac{1}{3}$ that passes through the point $(-3, 2)$?	**$y = (1/3)x + 3$**	$2 = (1/3)(-3) + b \rightarrow 2 = -1 + b \rightarrow b = 3$
A4	What is the slope of the line given by the equation $2x + 5y = 10$?	**$-2/5$**	$5y = -2x + 10 \rightarrow y = (-2/5)x + 2$
B1	What is the point of the line given by the point-slope equation $y + 2 = \frac{5}{4}(x - 8)$? Enter your answer as an ordered pair.	**$(8, -2)$**	$y + 2 = y - (-2)$: $x_1 = 8$, $y_1 = -2$.
B2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y + 5 = -2(x - 1)$**	**$y = -2x - 3$**	$y + 5 = -2x + 2 \rightarrow y = -2x - 3$
B3	What is the equation in slope-intercept form of a line with a slope of $\frac{3}{4}$ that passes through the point $(4, -1)$?	**$y = (3/4)x - 4$**	$-1 = (3/4)(4) + b \rightarrow -1 = 3 + b \rightarrow b = -4$
B4	What is the slope of the line given by the equation $4x - 2y = 8$?	**2**	$-2y = -4x + 8 \rightarrow y = 2x - 4$

C1	What is the point of the line given by the point-slope equation $y - 7 = -\frac{1}{2}(x + 3)$? Enter your answer as an ordered pair.	**(−3, 7)**	$x + 3 = x - (-3)$: $x_1 = -3$, $y_1 = 7$.
C2	Rewrite the equation in slope-intercept form. Simplify your answer. **$y - 6 = 5(x + 2)$**	**y = 5x + 16**	$y - 6 = 5x + 10 \rightarrow y = 5x + 16$
C3	What is the equation in slope-intercept form of a line with a slope of $-\frac{2}{5}$ that passes through the point (5, 1)?	**y = −(2/5)x + 3**	$1 = (-2/5)(5) + b \rightarrow 1 = -2 + b \rightarrow b = 3$
C4	What is the slope of the line given by the equation $6x + 3y = 9$?	**−2**	$3y = -6x + 9 \rightarrow y = -2x + 3$